Data Collection and Preprocessing Phase

Date	26 June 2026
Team ID	Individual
Project Title	Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues such as missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis of food ordering behaviour, consumer preferences, delivery performance, and revenue patterns across 50,000 records and 19 attributes.

Section	**Description**
Data Overview	The dataset contains 50,000 food ordering records with 19 columns: order_id, user_id, age, city, order_time, day_type, cuisine, meal_type, restaurant_type, order_value, discount_applied, delivery_fee, time_taken_to_order, rating_given, is_repeat_order, mood, hunger_level, company, and rainy_weather. It covers 6 cities (Mumbai, Pune, Bangalore, Delhi, Chandigarh, Hyderabad), 6 cuisine types, and 4 meal types. Order values range from ₹100 to ₹999 (avg ₹547.73) and delivery fees from ₹20 to ₹99 (avg ₹59.64). Customer ages span 18–44 years (avg ~31). All 19 columns are fully populated with zero null values and zero duplicate order IDs.
Data Cleaning	No missing values detected across all 19 columns. No duplicate order_id entries found (0 out of 50,000 records). Categorical columns (city, cuisine, meal_type, restaurant_type, mood,

	hunger_level, company, day_type, order_time) contain clean, consistent labels with no mixed-case or spelling variants. Boolean-style columns (discount_applied, is_repeat_order, rainy_weather) are uniformly encoded as Yes/No. Numerical columns (order_value, delivery_fee, age, rating_given, time_taken_to_order) contain no outliers beyond expected ranges. Dataset is confirmed clean and Tableau-ready.
Data Transformation	Filter records by city, cuisine, meal_type, and day_type for focused segment analysis. Sort by order_value descending to identify high-value orders and by rating_given to surface satisfaction trends. Pivot data to create city-wise and cuisine-wise aggregations of total order value and order count. Create calculated columns: Delivery Fee % = (delivery_fee / order_value) x 100; Order Value Tier (Low: ₹100–333, Mid: ₹334–666, High: ₹667–999); Ordering Speed Category based on time_taken_to_order (Fast: 1–4 min, Moderate: 5–9 min, Slow: 10–14 min).
Data Type Conversion	order_id and user_id confirmed as int64 — no conversion needed. order_value and delivery_fee confirmed as int64 — cast to float64 for ratio calculations in Tableau. age (int64, range 18–44) and time_taken_to_order (int64, range 1–14) are correctly typed. rating_given (int64, range 1–5) treated as discrete dimension in Tableau. All string columns (city, cuisine, meal_type, restaurant_type, mood, hunger_level, company, order_time, day_type, discount_applied, is_repeat_order, rainy_weather) confirmed as str and loaded as Tableau Dimensions. No type mismatches detected.
Column Splitting and Merging	No splitting required — all columns are already atomic and single-valued. Merge cuisine and meal_type into a Cuisine-Meal descriptor (e.g. Chinese – Dinner) for cross-segment heatmap analysis. Merge mood and hunger_level into a Behavioural Profile column (e.g. Stressed – High) to explore emotional-demand patterns. Combine day_type and order_time into a Session Label (e.g. Weekend – Evening) to identify peak ordering windows. Create a Customer Segment tag by combining restaurant_type and discount_applied (e.g. Premium – Discounted).
Data Modeling	Primary table: Orders (50,000 rows, 19 columns) used as single

	source of truth. Tableau calculated measures: Total Revenue = SUM(order_value); Average Order Value = AVG(order_value) = ₹547.73; Average Delivery Fee = AVG(delivery_fee) = ₹59.64; Average Rating = AVG(rating_given) = 2.99; Repeat Order Rate ≈ 49.97%; Discount Usage Rate ≈ 50.23%. City distribution is near-uniform (8,262–8,443 orders each). Cuisine distribution is near-uniform (8,228–8,452 each), with Desserts and Fast Food leading slightly.
Save Processed Data	Original dataset (food_ordering_behavior_dataset.csv, 50,000 rows x 19 columns) saved as primary source — no rows removed as the dataset is fully clean. Processed CSV with added calculated columns (Order Value Tier, Delivery Fee %, Ordering Speed Category, Cuisine-Meal, Behavioural Profile, Session Label) exported as food_ordering_processed.csv for Tableau ingestion. Tableau workbook saved as food_ordering_analysis.twbx with embedded data source. All files version-controlled in the GitHub repository with preprocessing steps documented in README.md.